In [1]:
```
import pandas as pd
import numpy as np
```

In [90]:
```
df=pd.read_csv(r"C:\Users\Ahmed akram\Downloads\social_media_posts( Data Cleaning ).csv")
```

In [91]:
```
df.head(5)
```

Out[91]:

	Index	Post_Date	Year	Month	City	Post_Type	Post_Details	Likes	Shares	Comments	Post_Link	Page
0	1	45746.0	2025	3	Beirut	Local News	Water supply disruption in Beirut.	1844.0	74	1553.0	https://example.com/post/1	ArabsNews
1	2	45569.0	2024	10	Beirut	Educational	Workshop on AI held in Beirut.	6661.0	2130	1329.0	https://example.com/post/2	ArabsNews
2	3	45648.0	2024	12	Alexandria	Political	Election updates from Alexandria.	3128.0	280	4885.0	https://example.com/post/3	ArabsNews
3	4	45428.0	2024	5	Dubai	Weather Alert	Storm warning issued in Dubai.	2758.0	750	3067.0	https://example.com/post/4	ArabsNews
4	5	45481.0	2024	7	Riyadh	Educational	Workshop on AI held in Riyadh.	1820.0	1387	2068.0	https://example.com/post/5	ArabsNews

In [92]:
```
df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 1012 entries, 0 to 1011
Data columns (total 12 columns):
 #   Column        Non-Null Count  Dtype
---  ------        --------------  -----
 0   Index         1012 non-null   int64
 1   Post_Date     1011 non-null   float64
 2   Year          1012 non-null   int64
 3   Month         1012 non-null   int64
 4   City          1012 non-null   str
 5   Post_Type     1012 non-null   str
 6   Post_Details  1011 non-null   str
 7   Likes         1011 non-null   float64
 8   Shares        1012 non-null   int64
 9   Comments      1010 non-null   float64
 10  Post_Link     1011 non-null   str
 11  Page          1012 non-null   str
dtypes: float64(3), int64(4), str(5)
memory usage: 180.5 KB
```

In [93]:
```
df[df.isnull().any(axis=1)]
```

Out[93]:

	Index	Post_Date	Year	Month	City	Post_Type	Post_Details	Likes	Shares	Comments	Post_Link	Page
137	138	45820.0	2025	6	Riyadh	Political	Election updates from Riyadh.	NaN	2630	4047.0	https://example.com/post/138	ArabsNews
211	212	45343.0	2024	2	Beirut	Religious	Charity campaign launched in Beirut.	313.0	263	NaN	https://example.com/post/212	ArabsNews
322	323	NaN	1900	1	Riyadh	Tech	NaN	7995.0	1486	NaN	NaN	ArabsNews

In [94]:
```
df.drop(322,inplace=True)
```

In [95]:
```
df[df.isnull().any(axis=1)]
```

Out[95]:

	Index	Post_Date	Year	Month	City	Post_Type	Post_Details	Likes	Shares	Comments	Post_Link	Page
137	138	45820.0	2025	6	Riyadh	Political	Election updates from Riyadh.	NaN	2630	4047.0	https://example.com/post/138	ArabsNews
211	212	45343.0	2024	2	Beirut	Religious	Charity campaign launched in Beirut.	313.0	263	NaN	https://example.com/post/212	ArabsNews

In [96]:
```
df['Likes'].max()
```

Out[96]:
```
np.float64(95760.0)
```

In [97]:
```
df.sort_values('Likes',ascending=False).head(3)
```

Out[97]:

	Index	Post_Date	Year	Month	City	Post_Type	Post_Details	Likes	Shares	Comments	Post_Link	Page
502	503	45778.0	2025	5	Alexandria	Local News	Water supply disruption in Alexandria.	95760.0	2917	995.0	https://example.com/post/503	ArabsNews
495	496	45530.0	2024	8	Baghdad	Educational	Scholarship applications open for Baghdad stud...	9993.0	2724	3856.0	https://example.com/post/496	ArabsNews
333	334	45770.0	2025	4	Jeddah	Weather Alert	Rainfall forecast updated.	9989.0	2407	1196.0	https://example.com/post/334	ArabsNews

In [98]:
```
df['Likes']=df['Likes'].replace(95760.0 ,9576.0)
```

In [99]:
```
df.sort_values('Likes',ascending=False).head(3)
```

Out[99]:

	Index	Post_Date	Year	Month	City	Post_Type	Post_Details	Likes	Shares	Comments	Post_Link	Page
495	496	45530.0	2024	8	Baghdad	Educational	Scholarship applications open for Baghdad stud...	9993.0	2724	3856.0	https://example.com/post/496	ArabsNews
333	334	45770.0	2025	4	Jeddah	Weather Alert	Rainfall forecast updated.	9989.0	2407	1196.0	https://example.com/post/334	ArabsNews
562	563	45628.0	2024	12	Beirut	Local News	New traffic rule enforced in Beirut.	9987.0	71	3258.0	https://example.com/post/563	ArabsNews

In [100…
```
df[df.isnull().any(axis=1)]
```

Out[100…

	Index	Post_Date	Year	Month	City	Post_Type	Post_Details	Likes	Shares	Comments	Post_Link	Page
137	138	45820.0	2025	6	Riyadh	Political	Election updates from Riyadh.	NaN	2630	4047.0	https://example.com/post/138	ArabsNews
211	212	45343.0	2024	2	Beirut	Religious	Charity campaign launched in Beirut.	313.0	263	NaN	https://example.com/post/212	ArabsNews

In [101…
```
df['Likes']=df['Likes'].fillna(df['Likes'].mean())
```

In [102…
```
df[df.isnull().any(axis=1)]
```

Out[102…

	Index	Post_Date	Year	Month	City	Post_Type	Post_Details	Likes	Shares	Comments	Post_Link	Page
211	212	45343.0	2024	2	Beirut	Religious	Charity campaign launched in Beirut.	313.0	263	NaN	https://example.com/post/212	ArabsNews

In [103…
```
df_filter=df[(df['Post_Type']=='Religious')&(df['City']=='Beirut')]
df_filter
```

Out[103...

	Index	Post_Date	Year	Month	City	Post_Type	Post_Details	Likes	Shares	Comments	Post_Link	Page
35	36	45436.0	2024	5	Beirut	Religious	Charity campaign launched in Beirut.	4127.0	69	2121.0	https://example.com/post/36	ArabsNews
72	73	45292.0	2024	1	Beirut	Religious	Charity campaign launched in Beirut.	6565.0	1448	3170.0	https://example.com/post/73	ArabsNews
96	97	45347.0	2024	2	Beirut	Religious	Event: Eid celebrations in Beirut.	9499.0	2160	3967.0	https://example.com/post/97	ArabsNews
117	118	45402.0	2024	4	Beirut	Religious	Event: Eid celebrations in Beirut.	4283.0	678	2282.0	https://example.com/post/118	ArabsNews
172	173	45648.0	2024	12	Beirut	Religious	Event: Eid celebrations in Beirut.	3292.0	2918	3042.0	https://example.com/post/173	ArabsNews
180	181	45768.0	2025	4	Beirut	Religious	Event: Eid celebrations in Beirut.	2478.0	144	3305.0	https://example.com/post/181	ArabsNews
199	200	45564.0	2024	9	Beirut	Religious	Charity campaign launched in Beirut.	7873.0	778	2189.0	https://example.com/post/200	ArabsNews
211	212	45343.0	2024	2	Beirut	Religious	Charity campaign launched in Beirut.	313.0	263	NaN	https://example.com/post/212	ArabsNews
245	246	45461.0	2024	6	Beirut	Religious	Charity campaign launched in Beirut.	3781.0	153	4179.0	https://example.com/post/246	ArabsNews
292	293	45454.0	2024	6	Beirut	Religious	Event: Eid celebrations in Beirut.	4694.0	2857	2759.0	https://example.com/post/293	ArabsNews
319	320	45506.0	2024	8	Beirut	Religious	Event: Eid celebrations in Beirut.	1230.0	1019	2547.0	https://example.com/post/320	ArabsNews
572	573	45535.0	2024	8	Beirut	Religious	Prayer times updated for Beirut.	4835.0	2511	1984.0	https://example.com/post/573	ArabsNews

	Index	Post_Date	Year	Month	City	Post_Type	Post_Details	Likes	Shares	Comments	Post_Link	Page
610	611	45822.0	2025	6	Beirut	Religious	Charity campaign launched in Beirut.	1811.0	142	451.0	https://example.com/post/611	ArabsNews
628	629	45740.0	2025	3	Beirut	Religious	Event: Eid celebrations in Beirut.	6617.0	377	4207.0	https://example.com/post/629	ArabsNews
725	726	45622.0	2024	11	Beirut	Religious	Prayer times updated for Beirut.	3405.0	572	3882.0	https://example.com/post/726	ArabsNews
794	795	45455.0	2024	6	Beirut	Religious	Event: Eid celebrations in Beirut.	386.0	1185	449.0	https://example.com/post/795	ArabsNews
880	881	45596.0	2024	10	Beirut	Religious	Prayer times updated for Beirut.	5448.0	1662	1114.0	https://example.com/post/881	ArabsNews
967	968	45321.0	2024	1	Beirut	Religious	Prayer times updated for Beirut.	8810.0	1902	1015.0	https://example.com/post/968	ArabsNews
970	971	45413.0	2024	5	Beirut	Religious	Charity campaign launched in Beirut.	6180.0	1579	4074.0	https://example.com/post/971	ArabsNews

In [106…
```
df['Comments']=df['Comments'].fillna(df_filter['Comments'].mean())
```

In [107…
```
df[df.isnull().any(axis=1)]
```

Out[107…

Index	Post_Date	Year	Month	City	Post_Type	Post_Details	Likes	Shares	Comments	Post_Link	Page

In [108…
```
df
```

Out[108...

	Index	Post_Date	Year	Month	City	Post_Type	Post_Details	Likes	Shares	Comments	Post_Link	Pag
0	1	45746.0	2025	3	Beirut	Local News	Water supply disruption in Beirut.	1844.0	74	1553.0	https://example.com/post/1	ArabsNew
1	2	45569.0	2024	10	Beirut	Educational	Workshop on AI held in Beirut.	6661.0	2130	1329.0	https://example.com/post/2	ArabsNew
2	3	45648.0	2024	12	Alexandria	Political	Election updates from Alexandria.	3128.0	280	4885.0	https://example.com/post/3	ArabsNew
3	4	45428.0	2024	5	Dubai	Weather Alert	Storm warning issued in Dubai.	2758.0	750	3067.0	https://example.com/post/4	ArabsNew
4	5	45481.0	2024	7	Riyadh	Educational	Workshop on AI held in Riyadh.	1820.0	1387	2068.0	https://example.com/post/5	ArabsNew
...	...	...	...	...	...	...	...	...	...	...	...	.
1007	996	45701.0	2025	2	Cairo	Tech	AI tech breakthrough reported.	3637.0	799	3933.0	https://example.com/post/996	ArabsNew
1008	997	45332.0	2024	2	Cairo	Local News	New traffic rule enforced in Cairo.	2474.0	2364	3675.0	https://example.com/post/997	ArabsNew
1009	998	45438.0	2024	5	Alexandria	Entertainment	Movie trailer released: 'Alexandria'.	234.0	1285	3924.0	https://example.com/post/998	ArabsNew
1010	999	45577.0	2024	10	Jeddah	Football	Match result: Dubai vs Dubai ended in a draw.	2116.0	2735	2084.0	https://example.com/post/999	ArabsNew
1011	1000	45684.0	2025	1	Amman	Entertainment	Movie trailer released: 'Amman'.	6444.0	1321	700.0	https://example.com/post/1000	ArabsNew

1011 rows × 12 columns

In [124…
```
df.loc[1007].to_frame()
```

Out[124…

	1007
Post_Date	45701.0
Year	2025
Month	2
City	Cairo
Post_Type	Tech
Post_Details	AI tech breakthrough reported.
Likes	3637.0
Shares	799
Comments	3933.0
Post_Link	https://example.com/post/996

In [122…
```
df=df.drop(['Index','Page'],axis=1)
```

In [123…
```
df
```

Out[123…

	Post_Date	Year	Month	City	Post_Type	Post_Details	Likes	Shares	Comments	Post_Link
0	45746.0	2025	3	Beirut	Local News	Water supply disruption in Beirut.	1844.0	74	1553.0	https://example.com/post/1
1	45569.0	2024	10	Beirut	Educational	Workshop on AI held in Beirut.	6661.0	2130	1329.0	https://example.com/post/2
2	45648.0	2024	12	Alexandria	Political	Election updates from Alexandria.	3128.0	280	4885.0	https://example.com/post/3
3	45428.0	2024	5	Dubai	Weather Alert	Storm warning issued in Dubai.	2758.0	750	3067.0	https://example.com/post/4
4	45481.0	2024	7	Riyadh	Educational	Workshop on AI held in Riyadh.	1820.0	1387	2068.0	https://example.com/post/5
...	...	...	...	...	...	...	...	...	...	...
1007	45701.0	2025	2	Cairo	Tech	AI tech breakthrough reported.	3637.0	799	3933.0	https://example.com/post/996
1008	45332.0	2024	2	Cairo	Local News	New traffic rule enforced in Cairo.	2474.0	2364	3675.0	https://example.com/post/997
1009	45438.0	2024	5	Alexandria	Entertainment	Movie trailer released: 'Alexandria'.	234.0	1285	3924.0	https://example.com/post/998
1010	45577.0	2024	10	Jeddah	Football	Match result: Dubai vs Dubai ended in a draw.	2116.0	2735	2084.0	https://example.com/post/999
1011	45684.0	2025	1	Amman	Entertainment	Movie trailer released: 'Amman'.	6444.0	1321	700.0	https://example.com/post/1000

1011 rows × 10 columns

In [126…

```
df.describe()
```

Out[126...

	Post_Date	Year	Month	Likes	Shares	Comments
count	1011.000000	1011.000000	1011.000000	1011.000000	1011.000000	1011.000000
mean	45560.362018	2024.320475	5.504451	4862.226733	1484.320475	2552.907517
std	157.496527	0.466890	3.403948	2821.598283	879.688203	1424.649874
min	45292.000000	2024.000000	1.000000	105.000000	12.000000	20.000000
25%	45424.000000	2024.000000	3.000000	2382.000000	730.500000	1340.500000
50%	45567.000000	2024.000000	5.000000	4756.000000	1421.000000	2547.000000
75%	45694.000000	2025.000000	8.000000	7204.500000	2263.000000	3785.500000
max	45832.000000	2025.000000	12.000000	9993.000000	3000.000000	4999.000000

In [131...
```
reach=df[['Likes','Shares','Comments']]
```

In [132...
```
reach
```

Out[132...

	Likes	Shares	Comments
0	1844.0	74	1553.0
1	6661.0	2130	1329.0
2	3128.0	280	4885.0
3	2758.0	750	3067.0
4	1820.0	1387	2068.0
...	...	...	...
1007	3637.0	799	3933.0
1008	2474.0	2364	3675.0
1009	234.0	1285	3924.0
1010	2116.0	2735	2084.0
1011	6444.0	1321	700.0

1011 rows × 3 columns

In [134...

```
reach.describe().astype(int)
```

Out[134...

	Likes	Shares	Comments
count	1011	1011	1011
mean	4862	1484	2552
std	2821	879	1424
min	105	12	20
25%	2382	730	1340
50%	4756	1421	2547
75%	7204	2263	3785
max	9993	3000	4999

In [135...

```
reach.corr()
```

Out[135...

	Likes	Shares	Comments
Likes	1.000000	0.003449	-0.029255
Shares	0.003449	1.000000	0.053781
Comments	-0.029255	0.053781	1.000000

In [140...

```
df['reach']=reach.sum(axis=1)
```

In [141...

```
df
```

Out[141...

	Post_Date	Year	Month	City	Post_Type	Post_Details	Likes	Shares	Comments	Post_Link	reach
0	45746.0	2025	3	Beirut	Local News	Water supply disruption in Beirut.	1844.0	74	1553.0	https://example.com/post/1	3471.0
1	45569.0	2024	10	Beirut	Educational	Workshop on AI held in Beirut.	6661.0	2130	1329.0	https://example.com/post/2	10120.0
2	45648.0	2024	12	Alexandria	Political	Election updates from Alexandria.	3128.0	280	4885.0	https://example.com/post/3	8293.0
3	45428.0	2024	5	Dubai	Weather Alert	Storm warning issued in Dubai.	2758.0	750	3067.0	https://example.com/post/4	6575.0
4	45481.0	2024	7	Riyadh	Educational	Workshop on AI held in Riyadh.	1820.0	1387	2068.0	https://example.com/post/5	5275.0
...	...	...	...	...	...	...	...	...	...	...	...
1007	45701.0	2025	2	Cairo	Tech	AI tech breakthrough reported.	3637.0	799	3933.0	https://example.com/post/996	8369.0
1008	45332.0	2024	2	Cairo	Local News	New traffic rule enforced in Cairo.	2474.0	2364	3675.0	https://example.com/post/997	8513.0
1009	45438.0	2024	5	Alexandria	Entertainment	Movie trailer released: 'Alexandria'.	234.0	1285	3924.0	https://example.com/post/998	5443.0
1010	45577.0	2024	10	Jeddah	Football	Match result: Dubai vs Dubai ended in a draw.	2116.0	2735	2084.0	https://example.com/post/999	6935.0
1011	45684.0	2025	1	Amman	Entertainment	Movie trailer released: 'Amman'.	6444.0	1321	700.0	https://example.com/post/1000	8465.0

1011 rows × 11 columns

In [145...

```
df.groupby('Post_Type')['reach'].mean().astype(int).sort_values(ascending=False).to_frame()
```

Out[145…

Post_Type	reach
Educational	9223
Religious	9132
Local News	9069
Political	9065
Entertainment	8962
Weather Alert	8774
Tech	8760
Health	8727
Football	8385

In [149…
```
df.to_excel(r"D:\New folder\analysis.xlsx",index=False)
```

In [150…
```
import matplotlib.pyplot as plt
```

In [151…
```
import seaborn as sb
```

In [153…
```
df.head(5)
```

Out[153...

	Post_Date	Year	Month	City	Post_Type	Post_Details	Likes	Shares	Comments	Post_Link	reach
0	45746.0	2025	3	Beirut	Local News	Water supply disruption in Beirut.	1844.0	74	1553.0	https://example.com/post/1	3471.0
1	45569.0	2024	10	Beirut	Educational	Workshop on AI held in Beirut.	6661.0	2130	1329.0	https://example.com/post/2	10120.0
2	45648.0	2024	12	Alexandria	Political	Election updates from Alexandria.	3128.0	280	4885.0	https://example.com/post/3	8293.0
3	45428.0	2024	5	Dubai	Weather Alert	Storm warning issued in Dubai.	2758.0	750	3067.0	https://example.com/post/4	6575.0
4	45481.0	2024	7	Riyadh	Educational	Workshop on AI held in Riyadh.	1820.0	1387	2068.0	https://example.com/post/5	5275.0

In [154...

```
reach
```

Out[154...

	Likes	Shares	Comments
0	1844.0	74	1553.0
1	6661.0	2130	1329.0
2	3128.0	280	4885.0
3	2758.0	750	3067.0
4	1820.0	1387	2068.0
...	...	...	...
1007	3637.0	799	3933.0
1008	2474.0	2364	3675.0
1009	234.0	1285	3924.0
1010	2116.0	2735	2084.0
1011	6444.0	1321	700.0

1011 rows × 3 columns

In [156...

```
df['reach'].to_frame
```

Out[156...

```
<bound method Series.to_frame of 0        3471.0
1       10120.0
2        8293.0
3        6575.0
4        5275.0
         ...
1007     8369.0
1008     8513.0
1009     5443.0
1010     6935.0
1011     8465.0
Name: reach, Length: 1011, dtype: float64>
```

In [173...

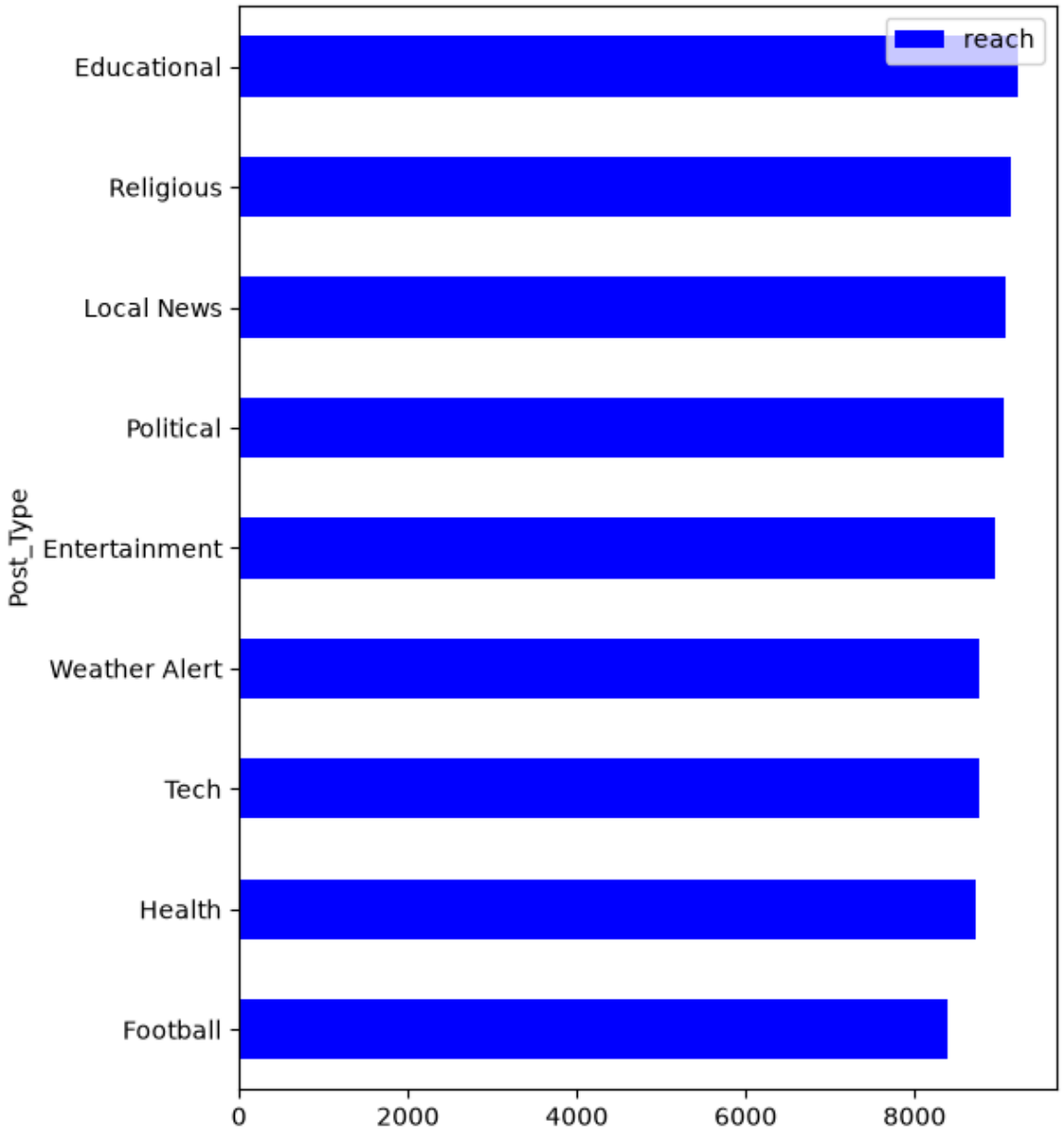
```
frt=df.groupby('Post_Type')['reach'].mean().astype(int).to_frame().sort_values('reach',ascending=True)
```

In [183…

```
frt.plot(kind='barh',figsize=(6,8),color='b')
```

Out[183…

```
<Axes: ylabel='Post_Type'>
```



In [176…

```
df['Post_Type'].unique()
```

Out[176…

```
<ArrowStringArray>
[   'Local News',   'Educational',     'Political', 'Weather Alert',
     'Football',     'Religious',          'Tech', 'Entertainment',
       'Health']
Length: 9, dtype: str
```

In [184…

```
df.head(5)
```

Out[184…

	Post_Date	Year	Month	City	Post_Type	Post_Details	Likes	Shares	Comments	Post_Link	reach
0	45746.0	2025	3	Beirut	Local News	Water supply disruption in Beirut.	1844.0	74	1553.0	https://example.com/post/1	3471.0
1	45569.0	2024	10	Beirut	Educational	Workshop on AI held in Beirut.	6661.0	2130	1329.0	https://example.com/post/2	10120.0
2	45648.0	2024	12	Alexandria	Political	Election updates from Alexandria.	3128.0	280	4885.0	https://example.com/post/3	8293.0
3	45428.0	2024	5	Dubai	Weather Alert	Storm warning issued in Dubai.	2758.0	750	3067.0	https://example.com/post/4	6575.0
4	45481.0	2024	7	Riyadh	Educational	Workshop on AI held in Riyadh.	1820.0	1387	2068.0	https://example.com/post/5	5275.0

In [188…

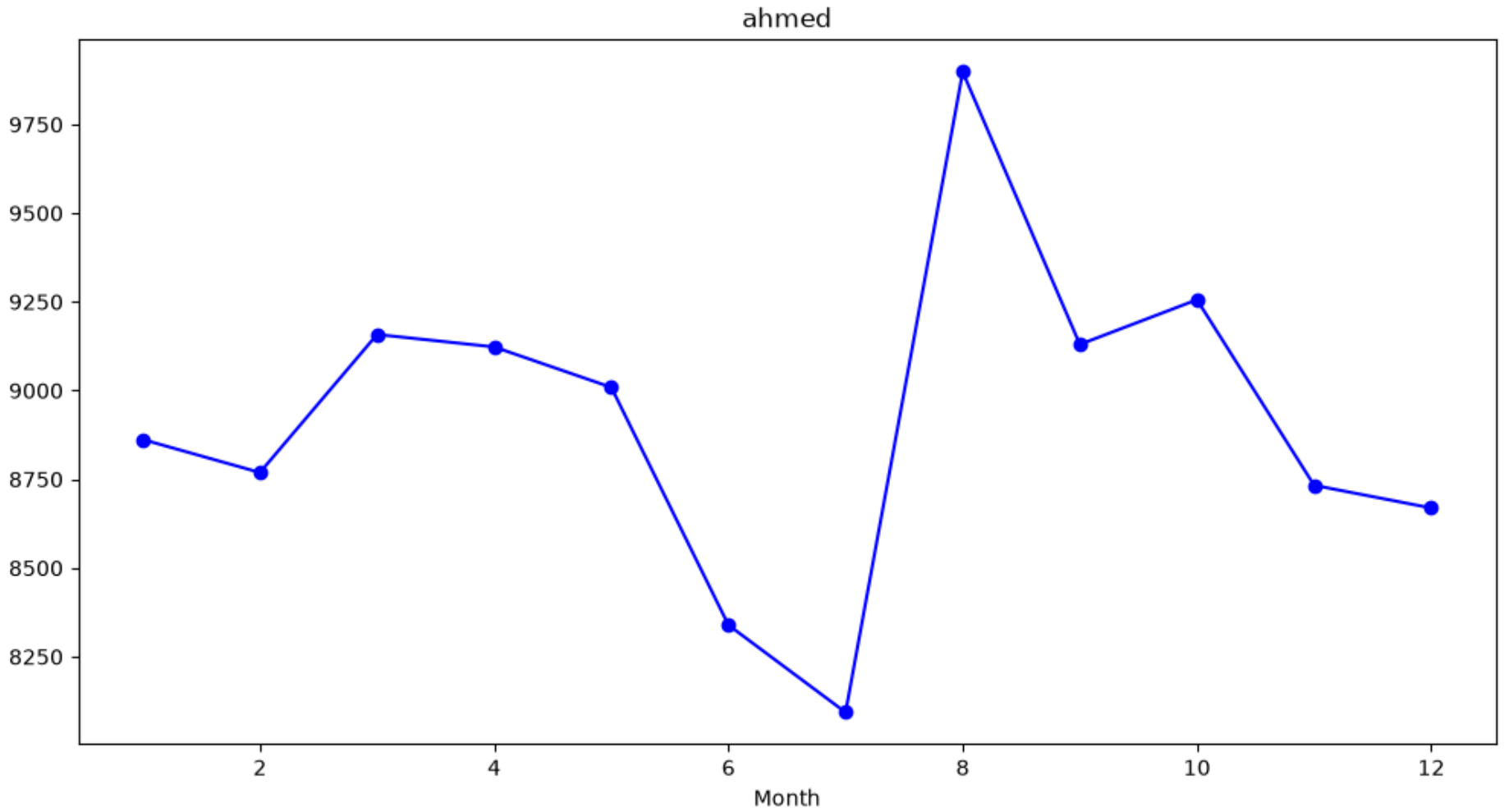
```
by_month=df.groupby('Month')['reach'].mean().astype(int)
by_month
```

Out[188…

```
Month
1     8862
2     8769
3     9158
4     9123
5     9009
6     8339
7     8093
8     9898
9     9130
10    9256
11    8733
12    8669
Name: reach, dtype: int64
```

In [193…
```
by_month.plot(kind='line',marker='o',color='b',title='ahmed',figsize=(12,6))
```

Out[193…
```
<Axes: title={'center': 'ahmed'}, xlabel='Month'>
```



In [194…
```
df['City'].unique()
```

Out[194…

```
<ArrowStringArray>
[    'Beirut', 'Alexandria',      'Dubai',     'Riyadh',     'Jeddah',
    'Baghdad',      'Cairo',      'Amman',       'Doha',       'dabi',
   '????????',       'doha', '??????????']
Length: 13, dtype: str
```

In [212…

```python
# 1. Filter selected cities
cities = ['Cairo', 'Dubai', 'Doha']
cap = df[df['City'].isin(cities)]

# 2. Calculate total reach for each city
reach_data = cap.groupby('City')['reach'].sum().sort_values(ascending=False)

# 3. Create professional Pie Chart
fig, ax = plt.subplots(figsize=(10, 7))

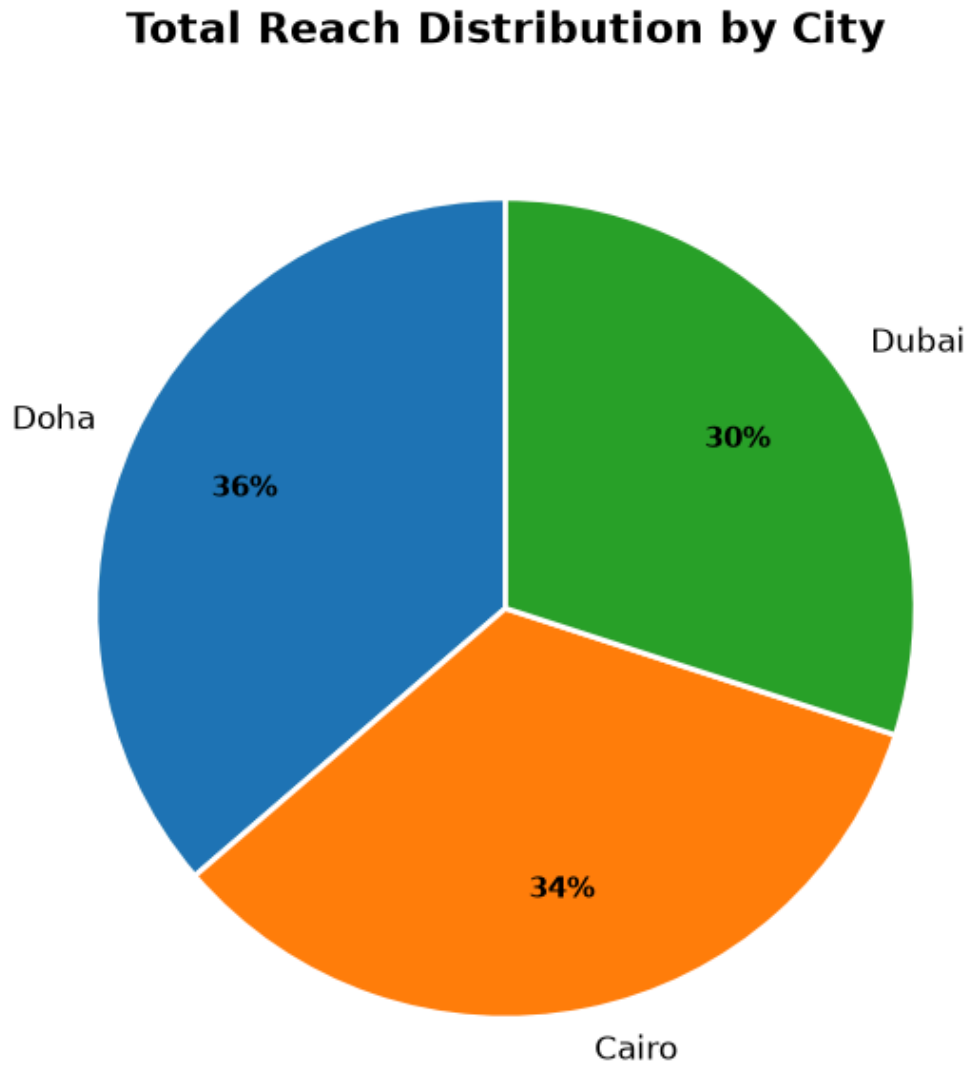
wedges, texts, autotexts = ax.pie(
    reach_data,
    labels=reach_data.index,
    autopct='%1.0f%%',
    startangle=90,
    pctdistance=0.7,
    wedgeprops={'edgecolor': 'white', 'linewidth': 2},
    textprops={'fontsize': 12}
)

# 4. Format percentage labels
for autotext in autotexts:
    autotext.set_fontsize(11)
    autotext.set_fontweight('bold')

# 5. Add title
ax.set_title(
    'Total Reach Distribution by City',
    fontsize=16,
    fontweight='bold',
    pad=20
)

# Keep the chart circular
```

```
plt.show()
```



In [ ]: